Prop $\mathbb{N}\times\mathbb{N} = \{(m,n) : m,n \in \mathbb{N}\}$ is countable.

Idea:

n \ m	1	2	3	...
1	$(1,1)$ 1	$(2,1)$ 3	$(3,1)$ 6	
2	$(1,2)$ 2	$(2,2)$ 5	$(3,2)$	
3	$(1,3)$ 4	$(2,3)$	$(3,3)$	
$\vdots$				

Pf: Set $f : \mathbb{N}\times\mathbb{N} \to \mathbb{N}$,

$$f(m,n) = \frac{(m+n-1)(m+n-2)}{2} + m.$$

Consider $g : \mathbb{N} \to \mathbb{N}\times\mathbb{N}$, $g(k) = (m, r+1-m)$, where:

① Pick $r \in \mathbb{N}$ s.t. $\frac{(r-1)r}{2} < k \leq \frac{r(r+1)}{2}$

(There is exactly one r that satisfies this.)

② Set $m = k - \frac{(r-1)r}{2}$.

(Idea:
- $r = m+n-1$ tells us which diagonal
- $m = k - \frac{(r-1)r}{2}$ tells us which entry in the diagonal.

The procedure/picture is more important to remember than the exact formula.)

Exc: $(f\circ g)(k) = k$ $\forall k \in \mathbb{N}$ and $(g\circ f)(m,n) = (m,n)$ $\forall (m,n) \in \mathbb{N}\times\mathbb{N}$.

Therefore f is bijective by HW1#6. $\square$

Cor The set of rational numbers

$$\mathbb{Q} = \{ \tfrac{a}{b} : a \in \mathbb{Z},\ b \in \mathbb{N} \}$$

is countable.

- "$b \in \mathbb{Z} \setminus \{0\}$" would also work. But for $b < 0$, we can absorb the minus sign into a

<u>Idea:</u>

$$\begin{array}{ccccc} \mathbb{Q} & \xrightarrow{\text{inj.}} & \overset{\mathbb{Z} \sim \mathbb{N}}{\mathbb{Z} \times \mathbb{N}} & \xleftrightarrow{\text{bij.}} & \mathbb{N} \times \mathbb{N} \\ \frac{a}{b} & & (a,b) & & \end{array}$$

(in lowest terms)

- So $\mathbb{Q}$ is an infinite subset of $\mathbb{N} \times \mathbb{N}$, which we know is countable
- Therefore $\mathbb{Q}$ is countable by Lecture 2.

<u>Next</u>: How to define addition and multiplication

- $+, \cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$: repeated addition, using (N2)
  $(m, n) \mapsto m+n$
  $m \cdot n$
- $+, \cdot : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$: extend to $-n$, $n \in \mathbb{N}$
- $+, \cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$?

<u>Problem</u>: Is this a function? There are many ways to write each rational number, e.g. $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \ldots$

EQUIVALENCE RELATIONS (Rudin Ch. 2)

<u>Def</u> An <u>equivalence relation</u> $\sim$ on a nonempty set A satisfies the following properties:

① reflexivity: $a \sim a \ \forall a \in A$.

② symmetry: if $a \sim b$ then $b \sim a$.

③ transitivity: if $a \sim b$ and $b \sim c$, then $a \sim c$.

<u>Ex</u> Given $a, b \in \mathbb{Z}$, define $a \sim b \iff a = b$. Then $\sim$ is an equivalence relation on $\mathbb{Z}$.

<u>Ex</u> Given sets A, B, define

$$A \sim B \iff A \text{ has the same cardinality as } B.$$

HW2: Prove that $\sim$ is an equivalence relation.

<u>Ex</u> Given $(a,b), (c,d) \in \mathbb{Z} \times \mathbb{N}$, define

$$(a,b) \sim (c,d) \iff ad = bc.$$

(Idea: $\frac{a}{b} = \frac{c}{d} \iff \underbrace{ad = bc}$.

This only requires multiplication on $\mathbb{Z}$!)

Prove that $\sim$ is an equivalence relation on $\mathbb{Z} \times \mathbb{N}$.

<u>Pf</u>: ① Reflexivity: Fix $(a,b) \in \mathbb{Z} \times \mathbb{N}$. Then $(a,b) \sim (a,b)$, since $ab = ba$.

② Symmetry: Suppose $(a,b) \sim (c,d)$. Then

$$(a,b) \sim (c,d) \Rightarrow ad = bc \Rightarrow cb = da$$
$$\Rightarrow (c,d) \sim (a,b).$$

③ Transitivity: Suppose $(a,b) \sim (c,d)$ and $(c,d) \sim (e,f)$. Then:

$$(a,b) \sim (c,d) \Rightarrow ad = \underline{bc} \Rightarrow adf = \underline{bcf}$$

$$(c,d) \sim (e,f) \Rightarrow \underline{cf} = de \Rightarrow \underline{bcf} = bde \Big\} \Rightarrow adf = bde$$

$$\Rightarrow \underbrace{d}_{\neq 0}(af - be) = 0 \Rightarrow af = be \Rightarrow (a,b) \sim (e,f). \quad \square$$

<u>Def</u> Let $\sim$ be an equivalence relation on a nonempty set A. The <u>equivalence class</u> of an element $a \in A$ is $C(a) = \{b \in A : b \sim a\}$.

<u>Ex</u> $a, b \in \mathbb{Z}$, $\quad a \sim b \iff a = b$.

The equivalence classes are $\ldots, \{-1\}, \{0\}, \{1\}, \ldots$

<u>Ex</u> A, B sets, $\quad A \sim B \iff$ same cardinality.

The equivalence classes are:

- Cardinality 0 sets
- Cardinality 1 sets
- Cardinality 2 sets

  $\vdots$

- Countable sets

  $\vdots$

<u>Ex</u> $(a,b), (c,d) \in \mathbb{Z} \times \mathbb{N}$, $\quad (a,b) \sim (c,d) \iff ad = bc$.

Then

$$C((1,2)) = \{(m,n) : \underbrace{(m,n) \sim (1,2)}_{\iff\ 2 \cdot m = 1 \cdot n}\}$$

$$= \{(m, 2m) : m \in \mathbb{N}\}$$

These are all the rational numbers $\frac{1}{2}, \frac{2}{4}, \frac{3}{6}, \ldots$

that are equal to $\frac{1}{2}$.

<u>Prop</u> (Properties of equivalence classes)

① $a \in C(a) \quad \forall a \in A$.

② If $a, b \in A$ satisfy $a \sim b$, then $C(a) = C(b)$.

③ If $a, b \in A$ do not satisfy $a \sim b$, then $C(a) \cap C(b) = \emptyset$.

④ $A = \bigcup_{a \in A} C(a)$.

<u>Pf</u>: ① Fix $a \in A$. Then $a \sim a$ by reflexivity, and so $a \in C(a)$.

② Suppose $a \sim b$.

Claim: $C(a) \subseteq C(b)$. Let $c \in C(c)$. Then

$$c \in C(a) \Rightarrow \left.\begin{matrix} c \sim a \\ a \sim b \end{matrix}\right\} \underset{\text{transitivity}}{\Rightarrow} c \sim b \Rightarrow c \in C(b).$$

Similarly, $C(a) \supseteq C(b)$. Together, we conclude $C(a) = C(b)$.

③ Next time!